that the availability of lipids may regulate proliferative potential is also entering the clinical arena where ongoing studies are evaluating the efficacy of lipid-lowering statins as anti-cancer drugs [14,20]. These intriguing new observations by Miettinen *et al.* [10] reveal that the molecular and genetic pathways that balance cell growth with proliferation are perhaps even more complicated than previously suspected.

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Neurobiology: Sensory Lateralization in the Fish Brain

In zebrafish, the dorsal habenula shows conspicuous left-right differences. New research shows that the left and right habenula differentially process visual and olfactory information. Spontaneous activity in habenular circuits may lead to activation of distinct neuronal targets and behavioral programs.

Hitoshi Okamoto

The majority of the human population can control the right hand more skillfully than the left hand. Likewise, the left half of the human face is in most cases better at expressing a smile than the right half [1]. From such observations, it has been inferred that the left and right hemispheres of the human brain show functional differences. In the 19th century, Dax and Broca discovered a lesion in the left hemisphere of post-mortem brains in patients with severe impairment in speech ability [2,3], and subsequently, Wernicke found that a damage to another region of the left hemisphere was associated with an impairment of language comprehension [4]. These discoveries for the first time provided the evidence that human language ability is controlled by the two distinct areas of the left cortex. In the 20th century, by careful observation of the patients with a split brain in which the corpus callosum, the nerve fiber bundles connecting the left and right hemispheres reciprocally, was surgically severed, Sperry demonstrated that — by and large — the left hemisphere is specialized for logical thinking and language processing, while the right hemisphere is more adapted for shape recognition, or emotional and artistic functions [5]. It certainly might be a more efficient and universal strategy for expanding the capacity of the brain to have two hemispheres of the brain engaged in information processing of different categories rather than having them dedicated to redundantly same subjects. Indeed, functional asymmetry of the brain has been found in other animals, too. For example, the two hemispheres of birds display a complementary pattern of visual analysis. The left hemisphere is specialized for detailed object analysis, attends to local features and excels in the categorization of visual stimuli. In contrast, the right hemisphere extracts relational configurations of visual stimuli that can be relevant during spatial orientation [6]. Now, two reports by Dreosti *et al.* [7] and Jetti *et al.* [8] in this issue of *Current Biology* demonstrate that the left and right habenula in zebrafish have a distinct difference in the modality of sensory information which they process, *i.e.* the left and right dHb differentially process visual and olfactory information, respectively.

Although some areas of the human brain, such as the planum temporale [9], show anatomical asymmetry and several molecules such as LMO4 have been identified to be expressed asymmetrically in both hemispheres [10], it has been difficult to decisively attribute genes to the establishment of brain asymmetry only by studying humans. It is evident that genetically tractable model animals are required to understand the functional, anatomical and genetic basis of brain laterality. However, this has been difficult in mice,

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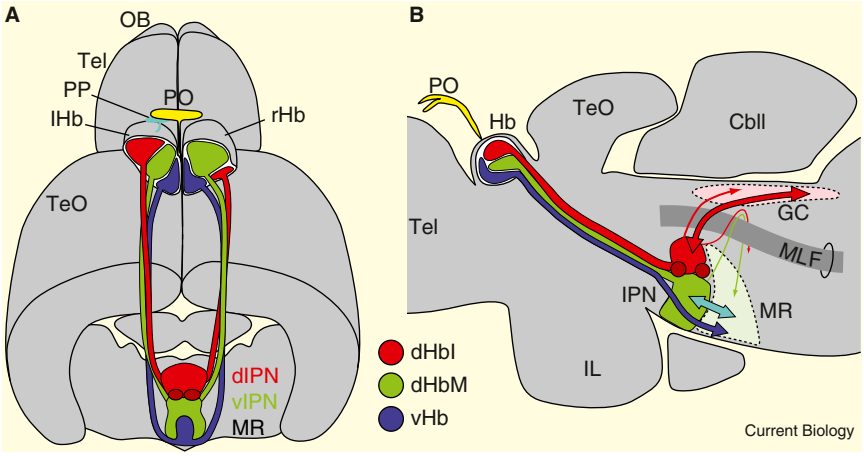


Figure 1. The habenulo-interpeduncular pathways in adult zebrafish.

(A) Dorsal oblique view; (B) sagittal view. The dHb in adult zebrafish can be subdivided into the medial (dHbM) and lateral subnuclei (dHbL). The dHbL neurons innervate the dIPN and iIPN, while the dHbM neurons innervate the vIPN and iIPN. These two subnuclei show a prominent left–right difference in the size ratio between the left and right dHb, *i.e.* the dHbL is larger than dHbM on the left and vice versa on the right side which accounts for the apparent asymmetrical neural connectivity between the dHb and the IPN.

Abbreviations: CbII, cerebellum; dHbL, lateral subnucleus of the dorsal habenula; dHbM, medial subnucleus of the dorsal habenula; dIPN, dorsal interpeduncular nucleus; GC, griseum centrale; IL, inferior lobe of the hypothalamus; lHb, left habenula; MLF, medial longitudinal fasciclus; MR, median raphe; olfactory bulb, olfactory bulb; TeO, optic tectum; PO, pineal organ; PP, parapineal organ; rHb, right habenula, Tel, telencephalon; vHb, ventral habenula; vIPN, ventral interpeduncular nucleus. Adapted from [14].

as the mouse cortex shows no apparent anatomical or functional laterality. Only recently, the CA3 neurons in the left and right hippocampus were found to show conspicuously asymmetric synaptic connection patterns with the apical and basal dendrites of CA1 neurons. But we are only beginning to understand the electrophysiological and behavioral significance and the molecular mechanism for this subtle and complex laterality [11].

Another, more obvious example of left–right asymmetry was identified at the neural circuit level in the brain of zebrafish in the projection from the dorsal habenula (dHb) to its target, the interpeduncular nucleus (IPN). The habenula (Hb) is an evolutionarily conserved nucleus which occupies the dorsal-most part of the diencephalon and acts as the relay station to connect the limbic nuclei in the forebrain with various nuclei in the brain stem. In zebrafish, it was discovered that the left dHb is predominantly connected with the dorsal and intermediate parts of the IPN (dIPN, iIPN), while the right dHb is predominantly connected with the ventral and the intermediate parts of the IPN (vIPN, iIPN) (Figure 1A) [12,13]. However, while the molecular and genetic mechanisms for establishment of this asymmetric nerve connection pattern are quite well understood [14,15], its functional significance has been unclear.

By imaging neuronal activity of the dHb in 4-day-old zebrafish larvae, Dresosti *et al.* [7] have now for the first time clearly shown a striking asymmetry in the neural activity of the left and right dHb, *i.e.* the majority of cells with sensory responses are functionally lateralized. Most neurons that responded to a binocular light stimulus were located on the left dHb, while neurons that responded to an odor stimulus were located on the right dHb, with very few neurons responding to both modalities (Figure 2A). Consistent with the asymmetric connections of the left and right dHb with the dIPN and vIPN, light responses were also observed predominantly in the dIPN, whereas odor responses were localized to the vIPN (Figure 1A). To test this asymmetric sensory processing further, the authors used various manipulations to create ‘double-right’ and ‘double-left’ habenulae, in which most dHb neurons show the characters of either the right-sided neurons or the left-sided neurons. In double-right brains, odor responses were bilateral in the dHb and robust in the vIPN whereas light responses were severely reduced in both the dHb and dIPN. Conversely, in double-left brains, very few dHb neurons responded to odor whereas neurons on both left and right responded to light. Consequently, light-dependent responses in the dIPN were robust, whereas olfactory responses in the vIPN were almost absent. These results confirmed the lateralized processing of visual and olfactory information by the left and right dHb in zebrafish. Previous studies have revealed an asymmetric projection from the olfactory bulbs on both sides exclusively to the right dHb [16]. This asymmetric afferent projection may be responsible for the lateralized olfactory information processing by the right dHb.

In the other study, Jetti *et al.* [8] investigated the nature of olfactory information processing by the right dHb. In the olfactory bulb, each olfactory glomerulus is known to be highly tuned to respond only to the odors of a certain group (category) with similar chemical structures. Because of this, different odors induce activity patterns in the olfactory bulb which are highly category specific. This has led to the suggestion that the olfactory bulb circuits can categorize and discriminate odors. In contrast, the activity patterns observed in the dHb did not show such clear selectivity to different categories of odors. However, the authors also demonstrated that these activity patterns were not random at all. They were able to group neurons with similar function by assigning the neurons into an arbitrarily chosen number of clusters based on their odor responses, using a common clustering algorithm called ‘k-means clustering’ (Figure 2B), and discovered that those neurons belonging to the same cluster based on their odor response properties were also spatially clustered in the dHb. This suggested that there is a spatial map for olfactory processing in the dHb. To investigate how this map is generated, the authors observed the spontaneous activities of the dHb neurons when no olfactory stimuli were presented. Quite intriguingly, they were also not random but highly organized into what the authors call “functional clusters of neurons” that were activated simultaneously over time. Furthermore,

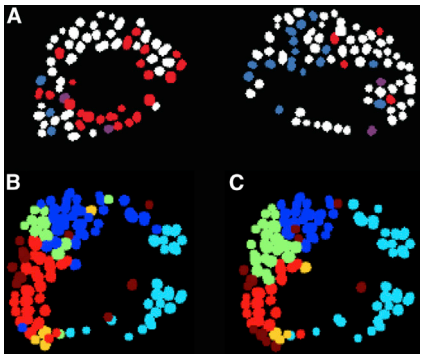


Figure 2. Spatial organization of odor and light responses in the dorsal habenula.

(A) Color-coded distribution of the dHb neuron cell bodies in the 4 day post-fertilization zebrafish larva which are activated differentially to the visual and olfactory stimuli with red, blue, violet, or white coding their responses to light, odour, both, or neither, respectively. Adapted from [7]. (B) Spontaneous and (C) food-odor evoked activity of dHb neurons are clustered using k-means clustering. Spatial positions of dHb neurons in each cluster are marked with different cluster colors. Adapted from [8].

the spatial location and the identity of functional dHb clusters during spontaneous activity and odor stimulation turned out to almost coincide in all individual cases of the recorded zebrafish (Figure 2B,C). These results suggested that dHb odor responses are governed by the spatially organized spontaneous activity of the dHb neurons.

What is the biological significance of these functional dHb clusters? The authors speculate that they may constitute attractor neural circuits that are composed of networks of recurrently connected neurons, whose dynamics show strong tendency to settle into defined stable states [17]. A previous study demonstrated that the dIPN is reciprocally connected with the dorsal tegmental area which might include the regulatory center for innately programed panic fear responses such as flight, fight and freeze, and the vIPN is reciprocally connected with the median raphe which might be involved in regulation of adaptive fear responses [18]. Furthermore, specific inactivation of neural transmission from the dHb into the dIPN converted the choice of fear response after classical fear conditioning of adult zebrafish from flight to freezing, suggesting that the dHb-IPN pathway is involved in the choice of innately encoded fear responses [18]. Based on these results, the authors have proposed the intriguing hypothesis that the odor stimulation in combination with the spontaneous dHb activity may attract the dHb neural ensembles to a particular 'internal' state, which could gate olfactory information toward the distinct dHb targets in the brain stem by way of the IPN to activate defined behavioral programs.

Another interesting issue is whether visual inputs are processed by the left dHb in a similar manner to how the olfactory inputs would be by the right dHb. Adult zebrafish are known to use preferentially the right eye when they are approaching novel objects [19]. In mutant zebrafish, in which the direction of the left–right asymmetry in the habenula was inverted, the left eye instead of the right eye was preferentially used for novelty recognition, suggesting the involvement of the dHb for this process [20]. It will be interesting to ask in the future whether visual stimulation by novel and familiar objects evokes activation of different functional clusters in the left dHb and to see if it might affect behavior or cognition accordingly.

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